

Customer Purchase & Sales Performance Analysis

Project Report

Data Cleaning • Analysis • Interactive Dashboard (Excel)

Customer Purchase & Sales Performance Analysis			
Total Sales Rev.	Transactions	Units Sold	Returning Sales
198,537	292	1,293	110,692
Top Category	Top City	New Sales Rev.	Avg. Sales
Electronics	Mogadishu	87,845	679.92

1. Executive Summary

- Cleaned 300 sales records; removed duplicates and errors.
- Built an interactive dashboard tracking revenue, customers, and categories.

2. Objectives

- Inspect and log every raw data quality issue.
- Clean identifiers, dates, categories, and numeric fields.
- Build a filterable KPI dashboard for quick insights.

3. Data Quality Issues Found

#	Data Quality Issue	Count
1	Duplicate Transaction_ID rows: 8 groups, 16 rows, 2 conflicting fields	16
2	Missing Transaction_ID values	15
3	Missing Purchase_Date values	15
4	Missing Customer_Type values	17
5	Null Customer_Type value	1
6	Missing Product_Category values	16
7	Invalid "-" Product_Category value	1
8	Missing Quantity values	16
9	Non-numeric "Unknown" Quantity values	3
10	Missing City values, incl. 1 "Unknown"	17
11	Missing Unit_Price values	16
12	Invalid Unit_Price: blank, N/A, negative	3

4. Data Cleaning Methodology

- Missing IDs replaced with traceable row-based placeholders.
- Duplicate rows reviewed; most complete record kept.
- Invalid values corrected, standardized, or flagged for review.

1	Transaction_ID	Purchase_Date	Customer_Type	Product_Category	Quantity	Unit_Price	City
2	TRX00114	2026-02-17	New	Groceries	2	\$19.50	Mogadishu
44	TRX00274	2025-02-11	New	Groceries	2	\$11.46	Hargeisa
93	TRX00217	2025-11-12	Returning	Home & Living	5	\$95.46	Mogadishu
99	TRX00092	2025-10-23	New	Groceries	6	\$25.06	Garowe
127	TRX00217	2025-11-12	Returning	Home & Living	5	\$95.46	Mogadishu
154	TRX00205	2025-05-04	New	Home & Living	5	\$73.55	Garowe
179	TRX00001	2025-09-20	Returning	Home & Living	2	\$135.55	Bosaso
192	TRX00098	2025-06-24	Returning	Clothing	3	\$46.32	Garowe
194	TRX00001	2025-09-20	Returning	Home & Living	2	\$135.55	Bosaso
204	TRX00205	2025-05-04	New	Home & Living	5	\$73.55	Garowe
209	TRX00114	2026-02-17	New	Groceries	2	\$19.50	Mogadishu
211	TRX00208	2025-10-17		Groceries	3	\$14.47	Kismayo
234	TRX00208	2025-10-17	New	Groceries		\$14.47	Kismayo
245	TRX00098	2025-06-24	Returning	Clothing	3	\$46.32	Garowe
281	TRX00092	2025-10-23	New	Groceries	6	\$25.06	Garowe
297	TRX00274	2025-02-11	New	Groceries	2	\$11.46	Hargeisa
302							

Fig. 1 — Duplicate Transaction_ID groups identified.

1	Missing ID'S				
2					
3	Product_Catego	Transaction_ID	Purchase_Date	NO	Solution
4	Clothing		2025-05-15	15 Missing Id's	Concatenate (Missing+row no)
5	Stationery		2025-03-15		
6	Electronics		2026-03-10		
16	Clothing		2025-05-29		
17	Groceries		2026-01-23		
18	Stationery		2025-09-03		
19					
20					
21	Missing Dates				
22					
23	Transaction_ID	Purchase_Date	Unit_Price	NO	
24	TRX00058		\$235.34	15 Missing Dates	
25	TRX00211		\$141.30		
26	TRX00210		\$39.08		
27	TRX00199		\$47.56		
38	TRX00223		\$263.68		
39					
40					

Fig. 2 — Missing IDs and dates resolved.

5. Dashboard: KPIs, Charts & Slicers

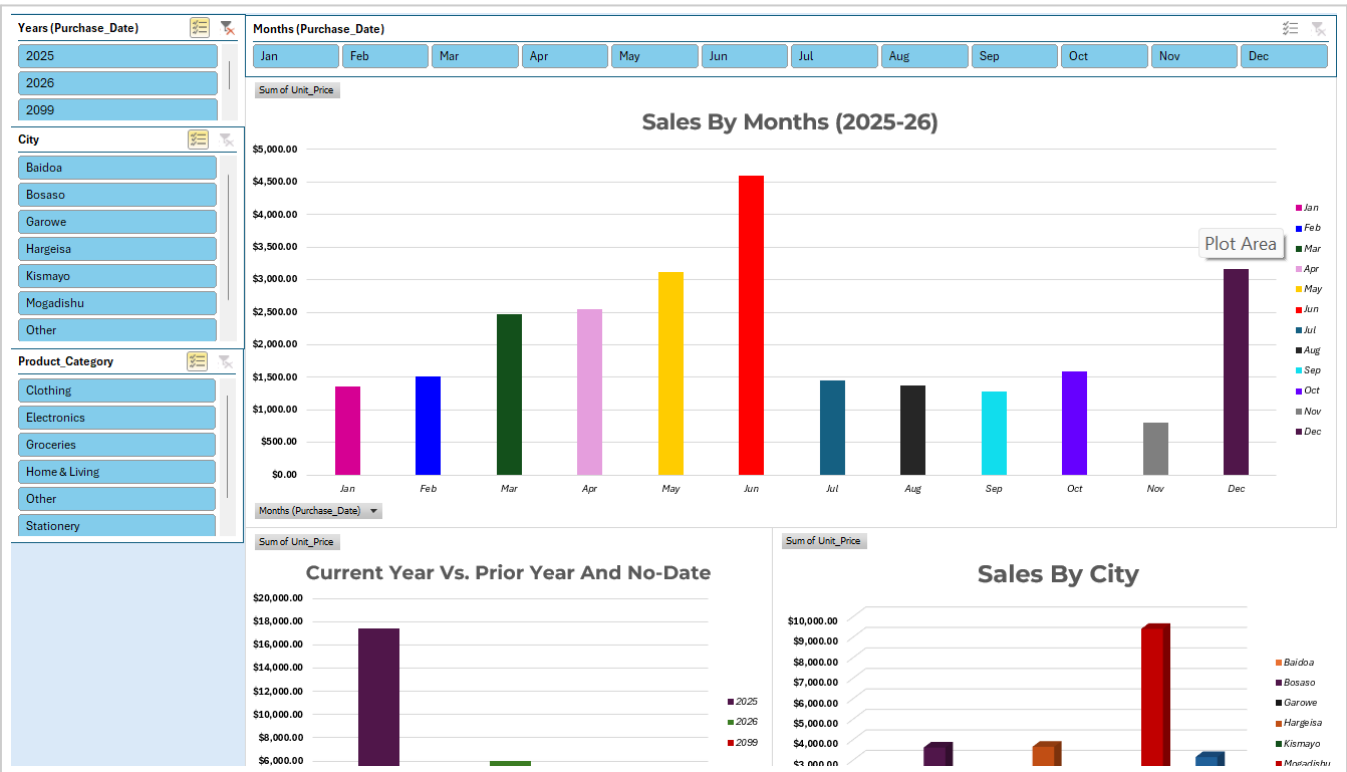


Fig. 3 — Interactive dashboard with slicers.

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6. Key Insights

- Electronics leads sales, four times the next category.
- Mogadishu leads all cities in tracked sales value.
- Returning customers nearly double new customer revenue.
- June peaked; November recorded the lowest sales.

7. Recommendations

- Correct invalid 2099-dated records from source documents.
- Standardize city and category entries at data entry.

8. Conclusion

- Raw data is now clean, organized, and dashboard-ready.